

Dorson Tang

📍 Richmond, BC ✉ dorsontang123@gmail.com 🌐 doruphin in Dorson Tang 🌐 Portfolio

Technical Skills

Languages: C/C++, Python, C#, JavaScript/TypeScript, Ruby, Java, Pascal

Tools/Frameworks: Linux, React, Git/Github, ROS 2, Gazebo, Unity, Tailwind, Django, PostgreSQL, Docker

Education

University of British Columbia

Bachelor of Science, Computer Science

Vancouver, BC

Expected May 2028

- **Related Coursework:** Data Structures and Algorithms, Networking, Computer Graphics, Relational Databases
- **Awards/Distinctions:** Dean's Scholar
- **GPA:** 4.00/4.33

Work Experience

Software Developer (C++)

Redbrick

Victoria, BC

May 2026 to Dec 2026

- Developed the backend of the Chromium-based Shift Browser, which is utilized by over **2,000,000** monthly active users
- Improved several internal CI/CD workflows, including overhauling the build system for x64 Macs by adding support for ARM cross-compiling, cutting daily build times by over **33%**
- Implemented front-end dialog with accompanying logic for vital data sync recovery paths

Software Developer

Ruboss

Vancouver, BC

May 2025 to Sept 2025

- Developed the latest iteration of a virtual e-book library from a legacy code base, utilizing various frameworks and technologies such as React Router, Tailwind CSS, and Ruby on Rails
- Created the new virtual course reader that is utilized by over **1,000,000** readers, as well as various custom React components used by over **10,000** authors with a focus on reusability and responsive UI design
- Optimized web embeds by **~66%** by byte size over network, allowing for lightweight implementation by authors

Software Team Lead

UBC Subbots

Vancouver, BC

Sept 2024 to Present

- Led **9** developers in the software sub-team of the UBC Subbots engineering design team, developing in a Linux Ubuntu environment and utilizing the ROS2 framework written in C++ and Python for autonomous robot control and simulation
- Implemented the VINS-Mono state estimator within AUV, allowing for robot localization in competition pool
- Architected a ROS2-to-Arduino serial communication service, enabling real-time control of hardware actuators
- Led software efforts to **semi-finals** in the international RoboSub 2025 competition with more than **50** schools in attendance

Projects

FretNot - C++, TypeScript, React, Tailwind

[Project Link](#) 🔗

- Created a guitar attachment in under 24 hours for Stormhacks 2025 that projects the proper fingering of chords using lasers, which won the **IEEE award** amongst **777** participants
- Developed the ESP32 code and the companion web interface, which allowed for the parsing of song lyrics and chords through the Tesseract OCR to sequentially play and project the chords alongside the song
- Extended Arduino library to incorporate modern C++ standards such as RAII principles and safe data types

Execution Order - C#, Unity

[Project Link](#) 🔗

- Ranked in the top **5%** amongst **37,000** participants in the largest game development competition in Itch.io history, the 2025 "Loop" Game Maker's Toolkit Game Jam
- Worked with art and design team to implement various visual effects such as VCR reversing and entity animations
- Implemented a robust window manager system which allowed for easy implementation of level obstacles through efficient use of several pillars of object-orientated programming such as encapsulation and polymorphism

